

Population Health Theories and Frameworks

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Chapter Overview

- Understanding why some people are healthier than others is a core focus of public health.
- Health is shaped by multiple factors, including:
 - Genetics
 - Lifestyle choices
 - Environment
 - Social conditions
- This presentation explores models explaining how determinants interact to influence population health.

What are determinants of health?

- **Determinants of health** are factors that influence an individual's or a population's health.
- These determinants are often categorized into **five key groups**.
- Understanding these determinants helps in designing effective health policies and interventions.

Biological and Genetic Determinants

- **Genetics:** Inherited traits that influence susceptibility to diseases (e.g., family history of diabetes or heart disease).
- **Biological Factors:** Age, sex, and genetic conditions (e.g., Down syndrome, cystic fibrosis).
- **Example:** A person with a genetic predisposition to high cholesterol may develop heart disease if environmental and behavioral factors also contribute.

Behavioral Determinants

- Individual **lifestyle choices** that impact health.
- Key behaviors include:
 - **Diet and Nutrition:** Poor diet leads to obesity and related diseases.
 - **Physical Activity:** Lack of exercise increases risk for cardiovascular diseases.
 - **Substance Use:** Tobacco, alcohol, and drug use significantly affect health outcomes.
- **Example:** Smoking is a leading cause of lung cancer and heart disease.

Social and Economic Determinants

- **Education and Income:** Higher levels of education correlate with better health outcomes.
- **Employment and Working Conditions:** Stable employment provides income and health benefits.
- **Social Support Networks:** Strong community ties improve mental and physical health.
- **Example:** Individuals in low-income communities often have limited access to healthy foods and healthcare services.

Environmental Determinants

- **Physical Environment:** Pollution, climate change, and access to clean water affect health.
- **Built Environment:** Availability of sidewalks, parks, and grocery stores.
- **Housing Conditions:** Overcrowding and poor sanitation increase disease risks.
- **Example:** High air pollution in urban areas is linked to increased rates of asthma and respiratory diseases.

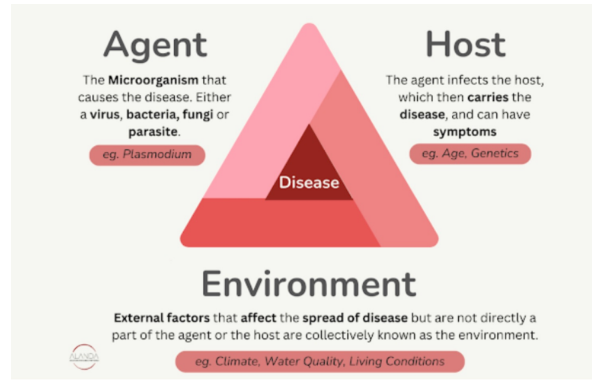
Healthcare Access and Quality

- **Availability of Healthcare Services:** Lack of access to preventive and emergency care worsens health disparities.
- **Health Insurance:** Affects affordability of medical care.
- **Cultural Competency:** Healthcare providers must address diverse patient needs.
- **Example:** Rural communities often experience healthcare shortages, leading to untreated conditions.

Why Do We Need Health Determinant Models?

- Health is not just about treating diseases—it's about **preventing** them.
- Important questions:
 - Why do wealthier neighborhoods have better health outcomes?
 - Why are chronic diseases more common in certain groups?
 - How do stress and social support affect mental health?
- Traditional **biomedical models** focus on treatment, but population health considers **broader social and environmental causes**.

1 The Epidemiologic Triad

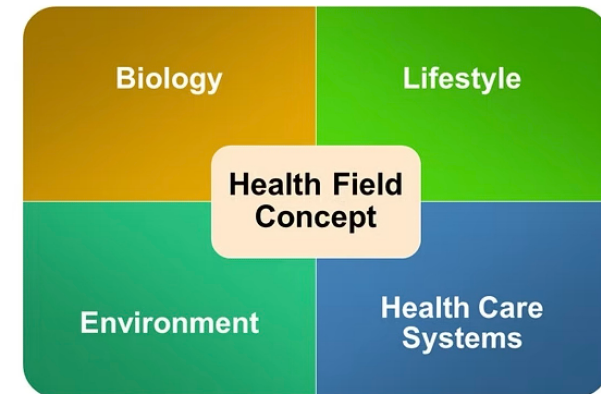


2 The Health Field Model

- A **basic model for infectious diseases** with three components:
 - Host** – Individual or population affected.
 - Agent** – Disease-causing factor (e.g., bacteria, viruses, smoking).
 - Environment** – External factors affecting exposure (e.g., air pollution, sanitation).
- Example: Tuberculosis**
 - Agent:** *Mycobacterium tuberculosis*
 - Host:** Weakened immunity due to malnutrition
 - Environment:** Overcrowding increases risk
- Limitation:** Does not fully explain chronic diseases.

- Shifted focus from infectious diseases to **chronic conditions**.
- Identifies **four major health determinants**:
 - Human Biology** – Genetic and physiological factors.
 - Lifestyle** – Behaviors like smoking, diet, exercise.
 - Environment** – Social and physical surroundings.
 - Healthcare System** – Access and quality of medical care.

2 The Health Field Model

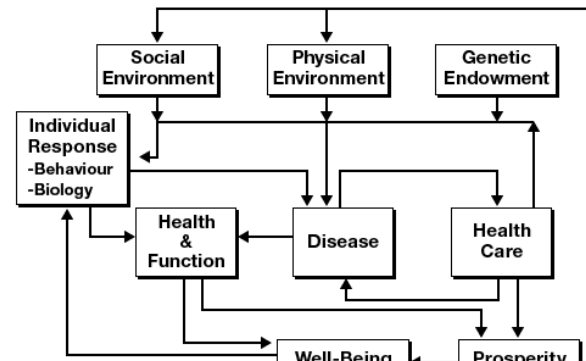


Example: Heart Disease

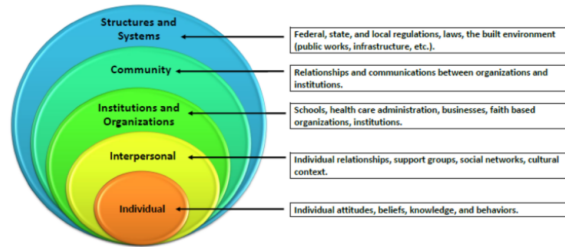
- Lifestyle:** Smoking, unhealthy diet.
- Environment:** Food deserts, limited access to healthy food.
- Healthcare System:** Focuses on treatment rather than prevention.
- Biology:** Family history of heart disease, genetic predisposition to high cholesterol

2 The Health Field Model

- A more **advanced model**



3 The Socio-ecological model



3 The Socio-ecological model

- Health is shaped at multiple levels:
 - Individual** – Genetics, behavior.
 - Interpersonal** – Family, social support.
 - Community** – Schools, workplaces, healthcare access.
 - Societal** – Laws, policies, economy.

Example: Obesity Prevention

- Individual:** Encouraging healthy diet.
- Community:** More grocery stores and parks.
- Societal:** Taxes on unhealthy foods.

4 The Life Course Perspective

- Health is **cumulative** and shaped over a lifetime
- Early-life exposures impact later health outcomes
- Critical periods: Childhood, adolescence, pregnancy
- Protective and risk factors evolve through different life stages

4 The Life Course Perspective

- Examines how **early-life experiences shape future health**.
- Key Concepts:**
 - Latency Effects** – Early-life exposures impact adult health.
 - Cumulative Effects** – Stress builds up over time.
 - Pathway Effects** – Disadvantages in childhood limit future opportunities.
- Example: Childhood Poverty & Health**
 - Poor nutrition in childhood → Increased risk for diabetes, heart disease in adulthood.
 - Chronic stress affects **brain development**.

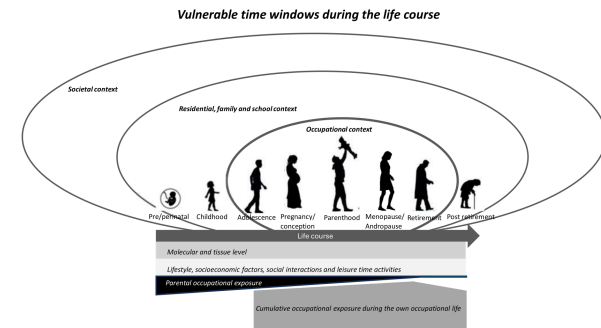
4 The Life Course Perspective

- Can geographic information keep you healthy? [youtube-link](#)
- Bill Davenport* from ESRI

5 Biopsychosocial Model

- Health is influenced by **biological, psychological, and social factors**
- Biological:** Genetics, immune function, neurobiology
- Psychological:** Stress, behavior, cognition
- Social:** Environment, socioeconomic status, relationships
- Interaction:** These domains interact dynamically, shaping health outcomes

4 The Life Course Perspective



4 The Life Course Perspective



5 Biopsychosocial Model

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